

Jain Koo

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RESEARCH INTEREST

- **Hardware–Software Co-design for AI Systems**
- **Efficient ML Systems and Computer Architecture**
- **Domain-Specific Accelerator Architectures and Dataflows**
- **Computer Architecture for Datacenter Efficiency**

EDUCATION

Master of Science | *Electrical & Electronic Engineering* Sep 2025 – Aug 2027
Yonsei University (Advisor : Prof. Won Woo Ro) Seoul, Korea

Bachelor of Science | *Electrical & Electronic Engineering, Earth System Science* Mar. 2021 – Aug 2025
Yonsei University Seoul, Korea

PUBLICATIONS

- [1] **Jain Koo**, Sungbin Kim, Enhyeok Jang, Sungwoo Kim, Yingyan (Celine) Lin, and Won Woo Ro, “**DiVA: Algorithm–System Co-Design for Visual Autoregressive Models via Submap-Level Retrievals**,” *IEEE International Symposium on High-Performance Computer Architecture (HPCA)*, In Submission, 2027
- [2] Hoje Lee, Minchan Kim, Jaewon Kwon, **Jain Koo**, Yongju Lee, Won Hur, Jiwon Lee, and Won Woo Ro, “**Maverick: Leveraging Address Translation to Mitigate Memory Latency in Server Workloads**,” *Proceedings of the IEEE/ACM International Symposium on Microarchitecture (MICRO)*, 2026.
- [3] **Jain Koo**, Enhyeok Jang, and Won Woo Ro, “Key–Value Cache Retrieval Is No Longer Cheap in Visual Autoregressive Models,” *The 41st International Technical Conference on Circuits/Systems, Computers and Communications (ITC-CSCC)*, 2026.

PROJECTS AND RESEARCH

Algorithm–System Co-design for Efficient Image Generation Jan 2025 – Present

- Worked in ISO 4 cleanroom developing applications to improve efficiency and creating specs
- Employed metrology and microscopy for failure analysis and developing process for wet etching
- Member of Emergency Response Team

Memory-Centric Architecture Using Reconfigurable PIM Devices Sep 2025 – Present

- Inventoried and maintained Physics Department lab equipment
- Physics tutoring

Development of PIM Software Architecture based on Data-Flow Computing Jan 2025 – Aug 2025

- Led a team of 20+ employees
- Ran social media, as well as all grassroots marketing

In-DRAM PIM-Based On-Device AI Acceleration Jan 2025 – Aug 2025

- Funded by Samsung Electronics
- Ran social media, as well as all grassroots marketing

RESEARCH EXPERIENCE

Graduate Student Researcher

Sep 2025 – Aug 2027

Embedded Systems and Computer Architecture Lab (eSCaL)

Yonsei University

- Worked in ISO 4 cleanroom developing applications to improve efficiency and creating specs
- Employed metrology and microscopy for failure analysis and developing process for wet etching
- Member of Emergency Response Team

Undergraduate Student Researcher

Jan 2025 – Sep 2025

Embedded Systems and Computer Architecture Lab (eSCaL)

Yonsei University

- Inventoried and maintained Physics Department lab equipment
- Physics tutoring

SCHOLARSHIPS AND AWARDS

BK (Brain Korea) 21 Research Fellowship

2025 – 2027

Graduate research fellowship funded through the BK21 program

Yonsei University

J.H. Foundation Research Fellowship

2026 – 2027

Merit-based scholarship for outstanding academic performance

J.H. Cultural Foundation

Graduate Student Research Assistant (GSRA) Scholarship

2025

Graduate research scholarship supporting research activities

Yonsei University

Academic Honors

2024

Recognition for outstanding academic performance

Yonsei University

The Yonsei Jinri Scholarship

2024

Merit-based scholarship for outstanding academic performance

Yonsei University

CONFERENCES AND PRESENTATIONS

Photometric Filter Fidelity and Use for Be Star Identification

November 2017

Austin College Physics Research Seminar

Reflectometry for Volumetric Soil Moisture Measurement

May 2017

Austin College Atmospheric Physics Fair

Design and Manufacturing of Products using 3D Printing

April 2017

Austin College Student Scholarship Conference

TEACHING EXPERIENCE

Electrical and Electronic Engineering Capstone Design (EEE4610) TA

Spring 2026

- Instructing groups of undergraduate students working on the Bachelors dissertation
- Directing research projects focused on accelerating Large Language Models and memory systems
- Hosting weekly meeting and advising the students on their research directions

COMMUNITY INVOLVEMENT

Austin College Community Tutors

Fall 2017 – Fall 2018

Free tutoring for local students in science and mathematics

Sherman, TX

River Legacy Nature Center

September 2015 – August 2016

Provided assistance for various youth science education programs

Arlington, TX

Back on My Feet Run Club

April 2014 – August 2015

Helping to reestablish homeless persons in the community

Austin, TX

SKILLS

Languages: Korean (Native), English (Fluent), Mandarin Chinese (Elementary), German (Elementary)

Programming: Python, C, C++, Verilog, CUDA, PyTorch